

Creation Date 28-Oct-2009

Revision Date 04-Oct-2023

Revision Number 7

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description: Hydrogen chloride, 1M solution in ethyl acetate
Cat No. : 388700000; 388701000
Molecular Formula Cl H

Unique Formula Identifier (UFI) 6H4M-MU3F-XW0N-GVA2

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Sector of use SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites
Product category PC21 - Laboratory chemicals
Process categories PROC15 - Use as a laboratory reagent
Environmental release category ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)
Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Company

EU entity/business name
Thermo Fisher Scientific
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

UK entity/business name
Fisher Scientific UK
Bishop Meadow Road,
Loughborough, Leicestershire LE11 5RG, United Kingdom

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
e-mail - infoch@thermofisher.com

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99
CHEMTREC Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

Poison Centre - Emergency information services

Ireland : National Poisons Information Centre (NPIC) -
01 809 2166 (8am-10pm, 7 days a week)
Malta : +356 2395 2000

SAFETY DATA SHEET

Hydrogen chloride, 1M solution in ethyl acetate

Revision Date 04-Oct-2023

Cyprus : +357 2240 5611

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Flammable liquids

Category 2 (H225)

Health hazards

Skin Corrosion/Irritation

Category 2 (H315)

Serious Eye Damage/Eye Irritation

Category 1 (H318)

Specific target organ toxicity - (single exposure)

Category 3 (H336)

Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Danger

Hazard Statements

H225 - Highly flammable liquid and vapor

H315 - Causes skin irritation

H318 - Causes serious eye damage

H336 - May cause drowsiness or dizziness

EUH066 - Repeated exposure may cause skin dryness or cracking

Precautionary Statements

P304 + P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

2.3. Other hazards

SAFETY DATA SHEET

Hydrogen chloride, 1M solution in ethyl acetate

Revision Date 04-Oct-2023

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Ethyl acetate	141-78-6	EEC No. 205-500-4	96	Flam. Liq. 2 (H225) Eye Irrit. 2 (H319) STOT SE 3 (H336) EUH066
Hydrochloric acid	7647-01-0	231-595-7	4	Skin Corr. 1A (H314) Eye Dam. 1 (H318) Acute Tox. 3 (H331) EUH071

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Hydrochloric acid	-	-	-

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.
Ingestion	Do NOT induce vomiting. Call a physician or poison control center immediately.
Inhalation	Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.
Self-Protection of the First Aider	Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Causes eye burns. . Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

SAFETY DATA SHEET

Hydrogen chloride, 1M solution in ethyl acetate

Revision Date 04-Oct-2023

5.1. Extinguishing media

Suitable Extinguishing Media

CO₂, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

Extinguishing media which must not be used for safety reasons

No information available.

5.2. Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂), Hydrogen chloride gas.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

6.2. Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance. Take precautionary measures against static discharges. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

SAFETY DATA SHEET

Hydrogen chloride, 1M solution in ethyl acetate

Revision Date 04-Oct-2023

7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Flammables area. Corrosives area.

Technical Rules for Hazardous Substances (TRGS) 510
Storage Class (LGK) (Germany)

Class 3

Switzerland - Storage of hazardous substances

Storage class - SC 3
<https://www.kvu.ch/de/themen/stoffe-und-produkte>
<https://www.kvu.ch/fr/themes/substances-et-produits>
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
Ethyl acetate	TWA: 734 mg/m ³ (15min) TWA: 200 ppm (15min) STEL: 1468 mg/m ³ (8h) STEL: 400 ppm (8h)	STEL: 1468 mg/m ³ 15 min STEL: 400 ppm 15 min TWA: 734 mg/m ³ 8 hr TWA: 200 ppm 8 hr	TWA / VME: 200 ppm (8 heures). TWA / VME: 734 mg/m ³ (8 heures). STEL / VLCT: 400 ppm. restrictive limit STEL / VLCT: 1468 mg/m ³ . restrictive limit	TWA: 200 ppm 8 uren TWA: 734 mg/m ³ 8 uren STEL: 400 ppm 15 minuten STEL: 1468 mg/m ³ 15 minuten	STEL / VLA-EC: 400 ppm (15 minutos). STEL / VLA-EC: 1468 mg/m ³ (15 minutos). TWA / VLA-ED: 200 ppm (8 horas) TWA / VLA-ED: 734 mg/m ³ (8 horas)
Hydrochloric acid	TWA: 5 ppm (8h) TWA: 8 mg/m ³ (8h) STEL: 10 ppm (15min) STEL: 15 mg/m ³ (15min)	STEL: 5 ppm 15 min STEL: 8 mg/m ³ 15 min TWA: 1 ppm 8 hr TWA: 2 mg/m ³ 8 hr	STEL / VLCT: 5 ppm. restrictive limit STEL / VLCT: 7.6 mg/m ³ . restrictive limit	TWA: 5 ppm 8 uren TWA: 8 mg/m ³ 8 uren STEL: 10 ppm 15 minuten STEL: 15 mg/m ³ 15 minuten	STEL / VLA-EC: 10 ppm (15 minutos). STEL / VLA-EC: 15 mg/m ³ (15 minutos). TWA / VLA-ED: 5 ppm (8 horas) TWA / VLA-ED: 7.6 mg/m ³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Ethyl acetate	TWA: 734 mg/m ³ 8 ore. Time Weighted Average TWA: 200 ppm 8 ore. Time Weighted Average STEL: 1468 mg/m ³ 15 minuti. Short-term STEL: 400 ppm 15 minuti. Short-term	TWA: 200 ppm (8 Stunden). AGW - exposure factor 2 TWA: 730 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 200 ppm (8 Stunden). MAK TWA: 750 mg/m ³ (8 Stunden). MAK Höhepunkt: 400 ppm Höhepunkt: 1500 mg/m ³	STEL: 1468 mg/m ³ 15 minutos STEL: 400 ppm 15 minutos TWA: 200 ppm 8 horas TWA: 734 mg/m ³ 8 horas	STEL: 1468 mg/m ³ 15 minuten TWA: 734 mg/m ³ 8 uren	TWA: 200 ppm 8 tunteina TWA: 730 mg/m ³ 8 tunteina STEL: 400 ppm 15 minuutteina STEL: 1470 mg/m ³ 15 minuutteina
Hydrochloric acid	TWA: 5 ppm 8 ore. Time Weighted Average TWA: 8 mg/m ³ 8 ore. Time Weighted Average	TWA: 2 ppm (8 Stunden). AGW - exposure factor 2 TWA: 3 mg/m ³ (8	STEL: 10 ppm 15 minutos STEL: 15 mg/m ³ 15 minutos	STEL: 15 mg/m ³ 15 minuten TWA: 8 mg/m ³ 8 uren	STEL: 5 ppm 15 minuutteina STEL: 7.6 mg/m ³ 15 minuutteina

SAFETY DATA SHEET

Hydrogen chloride, 1M solution in ethyl acetate

Revision Date 04-Oct-2023

	STEL: 10 ppm 15 minuti. Short-term STEL: 15 mg/m ³ 15 minuti. Short-term	Stunden). AGW - exposure factor 2 TWA: 2 ppm (8 Stunden). MAK TWA: 3.0 mg/m ³ (8 Stunden). MAK Höhepunkt: 4 ppm Höhepunkt: 6 mg/m ³	Ceiling: 2 ppm TWA: 5 ppm 8 horas TWA: 8 mg/m ³ 8 horas		
--	--	---	--	--	--

Component	Austria	Denmark	Switzerland	Poland	Norway
Ethyl acetate	MAK-KZGW: 400 ppm 15 Minuten MAK-KZGW: 1468 mg/m ³ 15 Minuten MAK-TMW: 200 ppm 8 Stunden MAK-TMW: 734 mg/m ³ 8 Stunden	TWA: 150 ppm 8 timer TWA: 540 mg/m ³ 8 timer STEL: 1468 mg/m ³ 15 minutter STEL: 400 ppm 15 minutter	STEL: 400 ppm 15 Minuten STEL: 1460 mg/m ³ 15 Minuten TWA: 200 ppm 8 Stunden TWA: 730 mg/m ³ 8 Stunden	STEL: 1468 mg/m ³ 15 minutach TWA: 734 mg/m ³ 8 godzinach	TWA: 200 ppm 8 timer TWA: 734 mg/m ³ 8 timer STEL: 400 ppm 15 minutter. value from the regulation STEL: 1468 mg/m ³ 15 minutter. value from the regulation
Hydrochloric acid	MAK-KZGW: 10 ppm 15 Minuten MAK-KZGW: 15 mg/m ³ 15 Minuten MAK-TMW: 5 ppm 8 Stunden MAK-TMW: 8 mg/m ³ 8 Stunden	STEL: 5 ppm 15 minutter STEL: 8 mg/m ³ 15 minutter	STEL: 4 ppm 15 Minuten STEL: 6 mg/m ³ 15 Minuten TWA: 2 ppm 8 Stunden TWA: 3 mg/m ³ 8 Stunden	STEL: 10 mg/m ³ 15 minutach TWA: 5 mg/m ³ 8 godzinach	Ceiling: 5 ppm Ceiling: 7 mg/m ³

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Ethyl acetate	TWA: 734 mg/m ³ TWA: 200 ppm STEL : 1468 mg/m ³ STEL : 400 ppm	TWA-GVI: 200 ppm 8 satima. TWA-GVI: 734 mg/m ³ 8 satima. STEL-KGVI: 400 ppm 15 minutama. STEL-KGVI: 1468 mg/m ³ 15 minutama.	TWA: 734 mg/m ³ 8 hr. TWA: 200 ppm 8 hr. STEL: 1468 mg/m ³ 15 min STEL: 400 ppm 15 min	STEL: 1468 mg/m ³ STEL: 400 ppm TWA: 734 mg/m ³ TWA: 200 ppm	TWA: 700 mg/m ³ 8 hodinách. Ceiling: 900 mg/m ³
Hydrochloric acid	TWA: 5 ppm TWA: 8.0 mg/m ³ STEL : 10 ppm STEL : 15.0 mg/m ³	TWA-GVI: 5 ppm 8 satima. TWA-GVI: 8 mg/m ³ 8 satima. STEL-KGVI: 10 ppm 15 minutama. STEL-KGVI: 15 mg/m ³ 15 minutama.	TWA: 8 mg/m ³ 8 hr. F TWA: 5 ppm 8 hr. STEL: 10 ppm 15 min STEL: 15 mg/m ³ 15 min	STEL: 10 ppm STEL: 15 mg/m ³ TWA: 5 ppm TWA: 8 mg/m ³	TWA: 8 mg/m ³ 8 hodinách. Ceiling: 15 mg/m ³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Ethyl acetate	TWA: 150 ppm 8 tundides. TWA: 500 mg/m ³ 8 tundides. STEL: 300 ppm 15 minutites. STEL: 1100 mg/m ³ 15 minutites.	TWA: 734 ppm 8 hr TWA: 200 mg/m ³ 8 hr STEL: 1468 ppm 15 min STEL: 400 mg/m ³ 15 min	STEL: 400 ppm STEL: 1468 mg/m ³ TWA: 200 ppm TWA: 734 mg/m ³	STEL: 1468 mg/m ³ 15 percekben. CK TWA: 734 mg/m ³ 8 órában. AK	TWA: 150 ppm 8 klukkustundum. TWA: 540 mg/m ³ 8 klukkustundum. Ceiling: 300 ppm Ceiling: 1080 mg/m ³
Hydrochloric acid	TWA: 5 ppm 8 tundides. TWA: 8 mg/m ³ 8 tundides. STEL: 10 ppm 15 minutites. STEL: 15 mg/m ³ 15 minutites.	TWA: 5 ppm 8 hr TWA: 8 mg/m ³ 8 hr STEL: 10 ppm 15 min STEL: 15 mg/m ³ 15 min	STEL: 5 ppm STEL: 7 mg/m ³ TWA: 5 ppm TWA: 7 mg/m ³	STEL: 16 mg/m ³ 15 percekben. CK TWA: 8 mg/m ³ 8 órában. AK	STEL: 5 ppm STEL: 8 mg/m ³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Ethyl acetate	STEL: 1468 mg/m ³ STEL: 400 ppm TWA: 200 mg/m ³ TWA: 54 ppm	Ceiling: 300 ppm Ceiling: 1100 mg/m ³ TWA: 150 ppm IPRD TWA: 500 mg/m ³ IPRD	TWA: 734 mg/m ³ 8 Stunden TWA: 200 ppm 8 Stunden STEL: 1468 mg/m ³ 15	TWA: 200 ppm TWA: 734 mg/m ³ STEL: 400 ppm 15 minuti STEL: 1468 mg/m ³ 15	TWA: 111 ppm 8 ore TWA: 400 mg/m ³ 8 ore STEL: 139 ppm 15 minute STEL: 500 mg/m ³ 15

SAFETY DATA SHEET

Hydrogen chloride, 1M solution in ethyl acetate

Revision Date 04-Oct-2023

			Minuten STEL: 400 ppm 15 Minuten	minuti	minute
Hydrochloric acid	STEL: 10 ppm STEL: 15 mg/m ³ TWA: 5 ppm TWA: 8 mg/m ³	TWA: 5 ppm IPRD TWA: 8 mg/m ³ IPRD STEL: 10 ppm STEL: 15 mg/m ³	TWA: 5 ppm 8 Stunden TWA: 8 mg/m ³ 8 Stunden STEL: 10 ppm 15 Minuten STEL: 15 mg/m ³ 15 Minuten	TWA: 5 ppm TWA: 8 mg/m ³ STEL: 10 ppm 15 minuti STEL: 15 mg/m ³ 15 minuti	TWA: 5 ppm 8 ore TWA: 8 mg/m ³ 8 ore STEL: 10 ppm 15 minute STEL: 15 mg/m ³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Ethyl acetate	TWA: 50 mg/m ³ 2417 MAC: 200 mg/m ³	Ceiling: 1100 mg/m ³ TWA: 200 ppm TWA: 734 mg/m ³	TWA: 200 ppm 8 urah TWA: 734 mg/m ³ 8 urah STEL: 400 ppm 15 minutah STEL: 1468 mg/m ³ 15 minutah	Binding STEL: 300 ppm 15 minuter Binding STEL: 1100 mg/m ³ 15 minuter TLV: 150 ppm 8 timmar. NGV TLV: 550 mg/m ³ 8 timmar. NGV	
Hydrochloric acid	MAC: 5 mg/m ³	Ceiling: 15 mg/m ³ TWA: 5 ppm TWA: 8.0 mg/m ³	TWA: 5 ppm 8 urah anhydrous TWA: 8 mg/m ³ 8 urah anhydrous STEL: 10 ppm 15 minutah anhydrous STEL: 15 mg/m ³ 15 minutah anhydrous	Binding STEL: 4 ppm 15 minuter Binding STEL: 6 mg/m ³ 15 minuter TLV: 2 ppm 8 timmar. NGV TLV: 3 mg/m ³ 8 timmar. NGV	TWA: 5 ppm 8 saat TWA: 8 mg/m ³ 8 saat STEL: 10 ppm 15 dakika STEL: 15 mg/m ³ 15 dakika

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Ethyl acetate 141-78-6 (96)				DNEL = 63mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Ethyl acetate 141-78-6 (96)	DNEL = 1468 mg/m ³ 400 ppm	DNEL = 1468 mg/m ³ 400 ppm	DNEL = 734 mg/m ³ 200 ppm	DNEL = 734mg/m ³
Hydrochloric acid 7647-01-0 (4)	DNEL = 15mg/m ³		DNEL = 8mg/m ³	

SAFETY DATA SHEET

Hydrogen chloride, 1M solution in ethyl acetate

Revision Date 04-Oct-2023

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Ethyl acetate 141-78-6 (96)	PNEC = 0.24mg/L	PNEC = 1.15mg/kg sediment dw	PNEC = 1.65mg/L	PNEC = 650mg/L	PNEC = 0.148mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Ethyl acetate 141-78-6 (96)	PNEC = 0.024mg/L	PNEC = 0.115mg/kg sediment dw		PNEC = 0.2g/kg food	

8.2. Exposure controls

Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection Goggles (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Butyl rubber	See manufacturers recommendations	-	EN 374	(minimum requirement)

Skin and body protection Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: Particulates filter conforming to EN 143 or Acid gases filter Type E Yellow conforming to EN14387

Small scale/Laboratory use Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.
Recommended half mask:- Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141
When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls No information available.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

SAFETY DATA SHEET

Hydrogen chloride, 1M solution in ethyl acetate

Revision Date 04-Oct-2023

Physical State	Liquid	
Appearance	Clear	
Odor	No information available	
Odor Threshold	No data available	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	No information available	
Flammability (liquid)	Highly flammable	On basis of test data
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Flash Point	-3 °C / 26.6 °F	Method - No information available
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
pH	No information available	
Viscosity	No data available	
Water Solubility	Insoluble	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
Ethyl acetate	0.73	
Vapor Pressure	No data available	
Density / Specific Gravity	0.900	
Bulk Density	Not applicable	Liquid
Vapor Density	No data available	(Air = 1.0)
Particle characteristics	(liquid) Not applicable	

9.2. Other information

Molecular Formula	Cl H
Molecular Weight	36.45
Explosive Properties	Vapors may form explosive mixtures with air

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Moisture sensitive.

10.3. Possibility of hazardous reactions

Hazardous Polymerization	Hazardous polymerization does not occur.
Hazardous Reactions	None under normal processing.

10.4. Conditions to avoid

Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. Exposure to moist air or water.

10.5. Incompatible materials

Strong oxidizing agents.

10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO₂). Hydrogen chloride gas.

SAFETY DATA SHEET

Hydrogen chloride, 1M solution in ethyl acetate

Revision Date 04-Oct-2023

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Ethyl acetate	10,200 mg/kg (Rat)	> 20 mL/kg (Rabbit) > 18000 mg/kg (Rabbit)	58 mg/l (rat; 8 h)
Hydrochloric acid	900 mg/kg (Rabbit)	> 5010 mg/kg (Rabbit)	LC50 = 4701 ppm (rat) 30 min (gas), LC50 = 588 ppm (4h) by extrapolation LC50 = 8.3 mg/L (rat) 30 min (aerosols) (MMAD < 5µm)

(b) skin corrosion/irritation;

Category 2

(c) serious eye damage/irritation;

Category 1

(d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

Component	Test method	Test species	Study result
Ethyl acetate 141-78-6 (96)	OECD Test Guideline 406	guinea pig	- non-sensitising

(e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

Component	Test method	Test species	Study result
Ethyl acetate 141-78-6 (96)	OECD Test Guideline 471 AMES test	in vitro Bacteria	negative
	OECD Test Guideline 473 Chromosomal aberration assay	in vitro Mammalian	negative
	OECD Test Guideline 476 Gene cell mutation	in vitro Mammalian	negative
	OECD Test Guideline 474 Mouse micronucleus assay	in vivo Mammalian	negative

(f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity;

Based on available data, the classification criteria are not met

Component	Test method	Test species / Duration	Study result
Ethyl acetate 141-78-6 (96)	OECD Test Guideline 416	Oral mouse	NOAEL = 26400

SAFETY DATA SHEET

Hydrogen chloride, 1M solution in ethyl acetate

Revision Date 04-Oct-2023

	OECD Test Guideline 414	2 Generation Inhalation Rat	mg/kg bw/day NOAEC = 73300 mg/m ³
--	-------------------------	-----------------------------------	--

(h) STOT-single exposure; Category 3

Results / Target organs Central nervous system (CNS).

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

Other Adverse Effects See actual entry in RTECS for complete information

Symptoms / effects, both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects . Do not empty into drains.

Component	Freshwater Fish	Water Flea	Freshwater Algae
Ethyl acetate	Fathead minnow: LC50: 230 mg/l/ 96h Gold orfe: LC50: 270 mg/L/48h	EC50 = 717 mg/L/48h	EC50 = 3300 mg/L/48h

Component	Microtox	M-Factor
Ethyl acetate	EC50 = 1180 mg/L 5 min EC50 = 1500 mg/L 15 min EC50 = 5870 mg/L 15 min EC50 = 7400 mg/L 2 h	

12.2. Persistence and degradability

Persistence Insoluble in water, Persistence is unlikely.

Component	Degradability
Ethyl acetate 141-78-6 (96)	79 % (20 d) (OECD 301 D)

12.3. Bioaccumulative potential

May have some potential to bioaccumulate

Component	log Pow	Bioconcentration factor (BCF)
Ethyl acetate	0.73	30 dimensionless

12.4. Mobility in soil

Spillage unlikely to penetrate soil The product is insoluble and floats on water . Is not likely mobile in the environment due its low water solubility.

SAFETY DATA SHEET

Hydrogen chloride, 1M solution in ethyl acetate

Revision Date 04-Oct-2023

12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

12.6. Endocrine disrupting properties

Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance
This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains.

Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number

UN2924

14.2. UN proper shipping name Technical Shipping Name

Flammable liquid, corrosive, n.o.s.
Ethyl acetate, Hydrogen chloride

14.3. Transport hazard class(es) Subsidiary Hazard Class

3
8

14.4. Packing group

II

ADR

14.1. UN number

UN2924

14.2. UN proper shipping name Technical Shipping Name

Flammable liquid, corrosive, n.o.s.
Ethyl acetate, Hydrogen chloride

14.3. Transport hazard class(es) Subsidiary Hazard Class

3
8

14.4. Packing group

II

IATA

14.1. UN number

UN2924

SAFETY DATA SHEET

Hydrogen chloride, 1M solution in ethyl acetate

Revision Date 04-Oct-2023

14.2. UN proper shipping name	Flammable liquid, corrosive, n.o.s.
Technical Shipping Name	Ethyl acetate, Hydrogen chloride
14.3. Transport hazard class(es)	3
Subsidiary Hazard Class	8
14.4. Packing group	II

14.5. Environmental hazards	No hazards identified
14.6. Special precautions for user	No special precautions required.
14.7. Maritime transport in bulk according to IMO instruments	Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Ethyl acetate	141-78-6	205-500-4	-	-	X	X	KE-00047	X	X
Hydrochloric acid	7647-01-0	231-595-7	-	-	X	X	KE-20189	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Ethyl acetate	141-78-6	X	ACTIVE	X	-	X	X	X
Hydrochloric acid	7647-01-0	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Ethyl acetate	141-78-6	-	Use restricted. See item 75. (see link for restriction details)	-
Hydrochloric acid	7647-01-0	-	Use restricted. See item 75. (see link for restriction details)	-

REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Ethyl acetate	141-78-6	Not applicable	Not applicable
Hydrochloric acid	7647-01-0	25 tonne	250 tonne

SAFETY DATA SHEET

Hydrogen chloride, 1M solution in ethyl acetate

Revision Date 04-Oct-2023

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification

Water endangering class = 1 (self classification)

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Ethyl acetate	WGK1	
Hydrochloric acid	WGK1	

Component	France - INRS (Tables of occupational diseases)
Ethyl acetate	Tableaux des maladies professionnelles (TMP) - RG 84

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Ethyl acetate 141-78-6 (96)		Group I	
Hydrochloric acid 7647-01-0 (4)	Prohibited and Restricted Substances		

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H225 - Highly flammable liquid and vapor
H314 - Causes severe skin burns and eye damage
H315 - Causes skin irritation
H318 - Causes serious eye damage
H319 - Causes serious eye irritation
H331 - Toxic if inhaled
H336 - May cause drowsiness or dizziness
EUH066 - Repeated exposure may cause skin dryness or cracking

Legend

SAFETY DATA SHEET

Hydrogen chloride, 1M solution in ethyl acetate

Revision Date 04-Oct-2023

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards On basis of test data

Health Hazards Calculation method

Environmental hazards Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Creation Date 28-Oct-2009

Revision Date 04-Oct-2023

Revision Summary Not applicable.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet